

MODULE 5

REGRESSION MODELS

Regression Models

- A *regression model* shows how a continuous value (the *dependent variable*) is related to a set of other values (*independent variables*)
- a regression model is used to predict values where they are unknown
- Sometimes, a regression model is simply used to explain phenomenon, but not to actually predict values
- Models can be used for many other purposes, including visualizing trends, analysis of variance tests, and testing variable significance

Simple Linear Models

- Regression analysis is a statistical technique for determining the relationship between two or more than two variables
- There are two types of variables in regression analysis
 - Independent variable
 - Dependent variable
- Independent variables are also known as predictor variables. These are the variables that do not change
- The variables whose values change are known as dependent variables
- Linear regression carry out a statistical procedure to predict the response variable based on the input of the independent or predictor variables.

Linear Regression

- Two types
- Simple Linear Regression –Based on the value of the single explanatory variable, the value of the response variable changes
- Multiple Linear Regression –The value is dependent upon more than one explanatory variables in case of multiple linear regression.

Simple Linear Regression

- Simple linear regression is used for finding the relationship between the dependent variable Y and the independent or predictor variable X.
- Both of these variables are continuous in nature

$$y_i = \beta_0 + \beta_1 x + \varepsilon$$

- y_i is calculated with the input variable X_i for every i^{th} data point
- The coefficients of regressions are denoted by β_0 and β_1
- The i^{th} value of x has ε_i as its error in the measurement

Multiple Linear Regression

- In many cases, there may be possibilities of dealing with more than one predictor variable for finding out the value of the response variable
- Therefore, the simple linear models cannot be utilised as there is a need for undertaking multiple linear regression for analyzing the predictor variables
- Using the two explanatory variables, we can delineate the equation of multiple linear regression as follows

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

- The two explanatory variables X_{1i} and X_{2i} , determine Y_i , for the i^{th} data point
- the predictor variables are also determined by the three parameters β_0 , β_1 , and β_2 of the model, and by the residual ϵ_i of the point i from the fitted surface.

Linear Regression - Example

- When you calculate the age of a child based on their height, you are assuming the older they are, the taller they will be
 - In this case, linear regression assumes that there exists a linear relationship between the response variable and the explanatory variables
 - This means that you can fit a line between the two (or more variables)
 - $\text{Height} = a + \text{Age} * b$
 - A linear regression can be calculated in R with the command **lm**
 - *$\text{lm}([\text{target}] \sim [\text{predictor} / \text{features}], \text{data} = [\text{data source}])$*

```
Call:
lm(formula = height ~ age, data = ageandheight)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27238 -0.24248 -0.02762  0.16014  0.47238

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  64.9283    0.5084  127.71  < 2e-16 ***
age           0.6350    0.0214   29.66 4.43e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.256 on 10 degrees of freedom
Multiple R-squared:  0.9888,    Adjusted R-squared:  0.9876
F-statistic: 880 on 1 and 10 DF,  p-value: 4.428e-11
```

- For Multiple Linear Regression

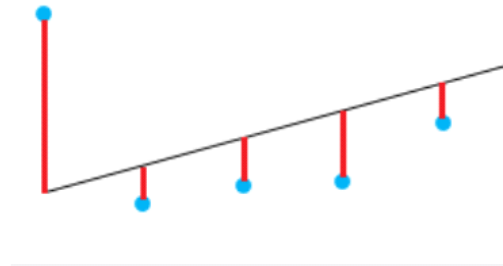
- $\text{Height} = a + \text{Age} \times b_1 + (\text{Number of Siblings}) \times b_2$
- You are now looking at the height as a function of the age in months and the number of siblings the child has
- `poly(x, ..., degree = 1, coefs = NULL, raw = FALSE)`
- the `poly` function takes a vector `x` (or a set of vectors), `degree` to specify a maximum degree to generate, `coefs` to specify coefficients from a previous fit (when using `poly` to generate predicted values), and `raw` to specify whether to use raw and not orthogonal polynomials

1. Coefficients

- In the red square, you can see the values of the intercept (“a” value) and the slope (“b” value) for the age
- These “a” and “b” values plot a line between all the points of the data
- So, in this case, if there is a child that is 20.5 months old, a is 64.92, and b is 0.635, the model predicts (on average) that its height in centimeters is around $64.92 + (0.635 * 20.5) = 77.93$ cm.

2. Residuals

- A good way to test the quality of the fit of the model is to look at the residuals or the differences between the real values and the predicted values
- The straight line in the image above represents the predicted values.
- The red vertical line from the straight line to the observed data value is the residual.



- P value

```
call:
lm(formula = height ~ age + no_siblings, data = ageandheight)

Residuals:
    Min       1Q   Median       3Q      Max
-0.28029 -0.22490 -0.02219  0.14418  0.48350

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  64.95872    0.55752  116.515 1.28e-15 ***
age          0.63516    0.02254   28.180 4.34e-10 ***
no_siblings -0.01137    0.05893   -0.193  0.851
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2693 on 9 degrees of freedom
Multiple R-squared:  0.9888,    Adjusted R-squared:  0.9863
F-statistic: 397.7 on 2 and 9 DF, p-value: 1.658e-09
```

- A standard way to test if the predictors are not meaningful is by looking if the p-values smaller than 0.05.
- The p-value for age is 4.34×10^{-10} or 0.0000000000434. A very small value means that age is probably an excellent addition to your model.
- The p-value for the number of siblings is 0.85. In other words, there's an 85% chance that this predictor is not meaningful for the regression

Predict the Values Using a Model

- To get the vector of residuals from a linear model fit, use the residuals function: **residuals(object, ...)**
- To get a vector of fitted values, use the fitted function: **fitted(object, ...)**
- use the predict function to calculate predicted values using the model object and another data frame:

predict(object, newdata, se.fit = FALSE, scale = NULL, df = Inf, interval = c("none", "confidence", "prediction"), level = 0.95, type = c("response", "terms"), terms = NULL, na.action = na.pass, pred.var = res.var/weights, weights = 1, ...)

- object specifies the model returned by the fitting function, newdata specifies a new data source for predictions, and na.action specifies how to deal with missing values in newdata.

Analyzing the fit

- One measure very used to test how good your model is the coefficient of determination or R^2 .
- This measure is defined by the proportion of the total variability explained by the regression model

$$R^2 = \frac{\text{Explained Variation of the model}}{\text{Total variation of the mode}}$$

- For models that fit the data well, R^2 is near 1. Models that poorly fit the data have R^2 near 0. In the examples below, the first one has an R^2 of 0.02; this means that the model explains only 2% of the data variability
- The second one has an R^2 of 0.99, and the model can explain 99% of the total variability

Refining the Model

- it is better to use the update function to refit a model
- You can run update after changing the formula (perhaps adding or subtracting a term) or even after changing the data frame.
- ```
> runs.mdl2 <- update(runs.mdl, formula=runs ~ singles + doubles +
+ triples + homeruns + walks + hitbypitch +
+ stolenbases + caughtstealing + 0)
```
- ```
> runs.mdl2
```

Nonlinear Least Square

- To fit nonlinear models (minimizing least squares error), you can use the `nls` function:

```
nls(formula, data, start, control, algorithm,  
    trace, subset, weights, na.action, model,  
    lower, upper, ...)
```

Argument	Description	Argument	Description
formula	A formula object that specifies the form of the model to fit.	subset	An optional vector specifying the set of rows to include.
data	A data frame in which formula can be evaluated.	weights	An optional vector specifying weights for observations.
start	A named list or named vector with starting estimates for the fit.	na.action	A function that specifies how to treat NA values in the data.
control	A list of arguments to pass to control the fitting process (see the help file for <code>nls.control</code> for more information).	model	A logical value specifying whether to include the model frame as part of the model object.
algorithm	The algorithm to use for fitting the model. Use <code>algorithm="plinear"</code> for the Golub-Pereyra algorithm for partially linear least squares models and <code>algorithm="port"</code> for the 'nl2sol' algorithm from the Port library.	lower	An optional vector specifying lower bounds for the parameters of the model.
trace	A logical value specifying whether to print the progress of the algorithm while <code>nls</code> is running.	upper	An optional vector specifying upper bounds for the parameters of the model.
		...	Additional arguments (not currently used).